Lecture 28: Markov chain Monte Carlo (MCMC)

- Lots of "gadgets" in scientific research: telescopes, microscopes...seems (relatively) simple now but they were sophisticated at the time. Data coming out of particle colliders are extremely complex to model.
- These complex data is only useful with parallel computational power, scientific computing paradigms and data analysis tools.



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Drawing the Bayesian model (owl)

When is MCMC needed?

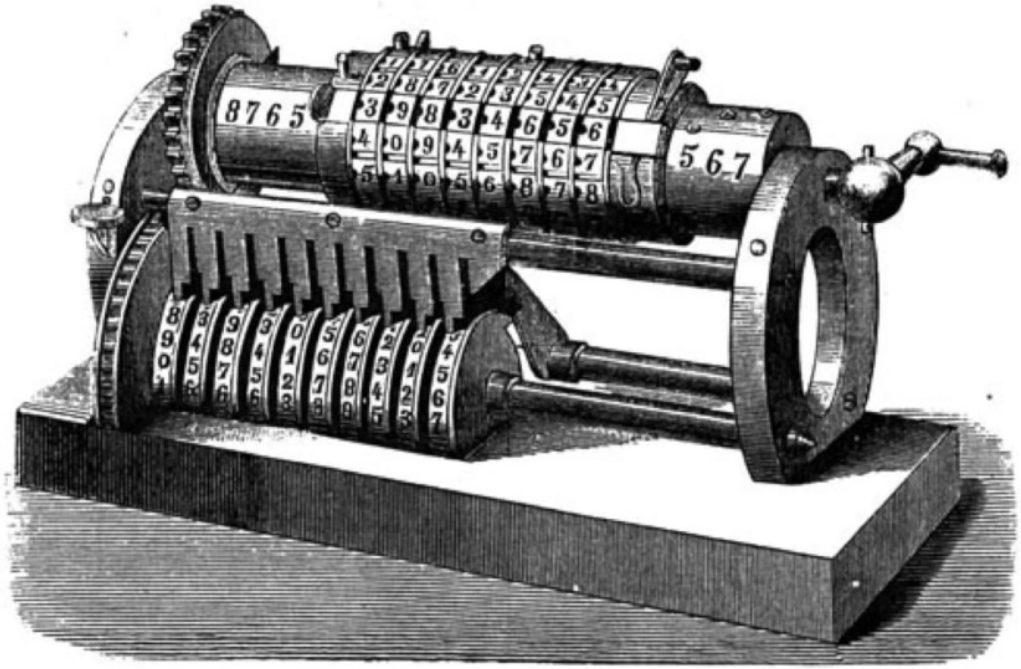
1. Theoretical estimand
2. Scientific (causal) model(s)
3. Use 1. & 2, to build statistical model(s)
4. Simulate from 2. to validate 3. yields a procedure to get 1.
5. Analyse real data



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Introduction to what is happening in the box for posterior estimation

How to compute the posterior?

- Analytical approach; often impossible
- Grid approximation; very intensive and sometimes grid resolution an issue
- Quadratic (or Laplace) approximation; limited to cases where the posterior is approximately Gaussian

NEW Markov chain Monte Carlo; intensive



Good King Markov and his kingdom if islands

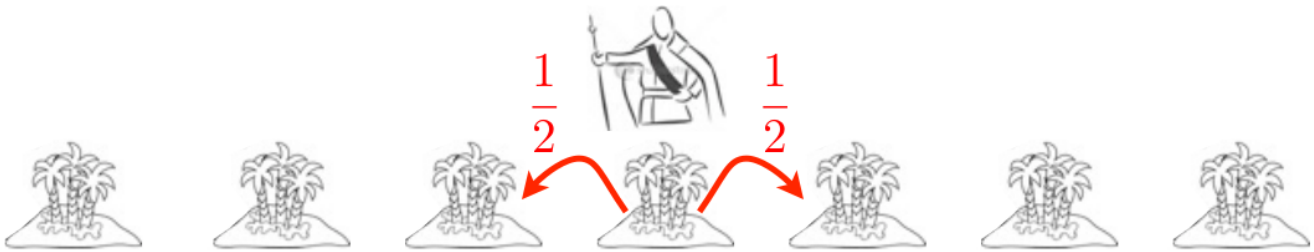
The Metropolis archipelago



- King Markov has to visit each island in proportion to its population size.
- He wants an easy rule scheme, without long records.
- Here is how he does it.

The Metropolis archipelago

Fair visit algorithm

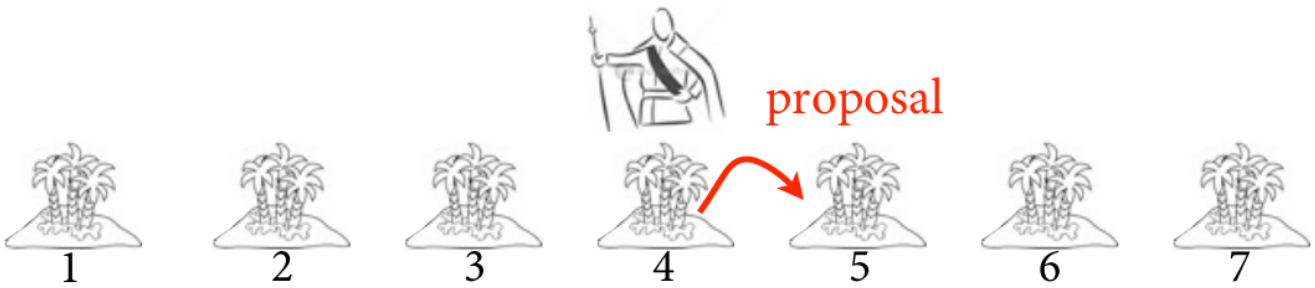


1. Flip a coin to choose an island left or right. Call it the "proposal" island.

The Metropolis archipelago

Fair visit algorithm

1. Flip a coin to choose an island left or right.
   Call it the "proposal" island.

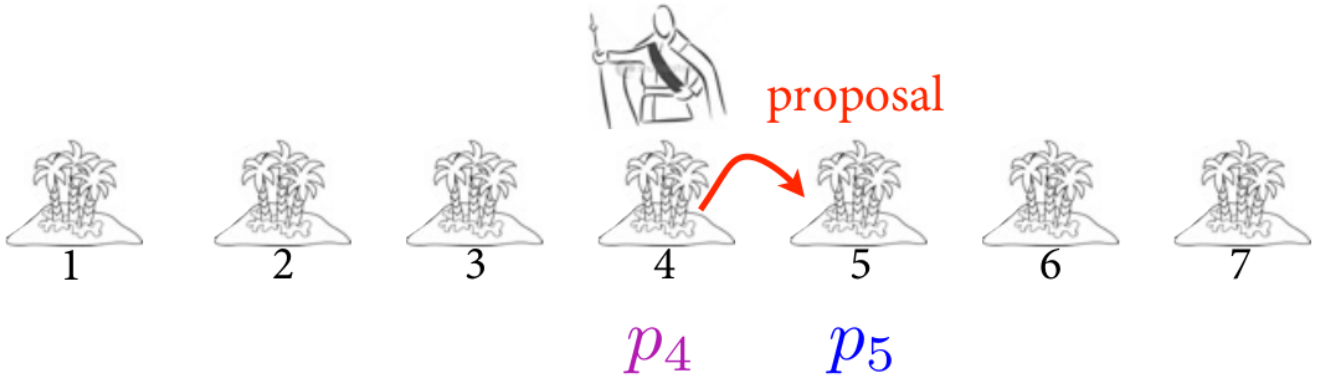


2. Find the population of the proposal island (p_5 here).

The Metropolis archipelago

Fair visit algorithm

1. small Flip a coin to choose an island left or right. Call it the "proposal" island.
2. Find the population of the proposal island (p_5 here).

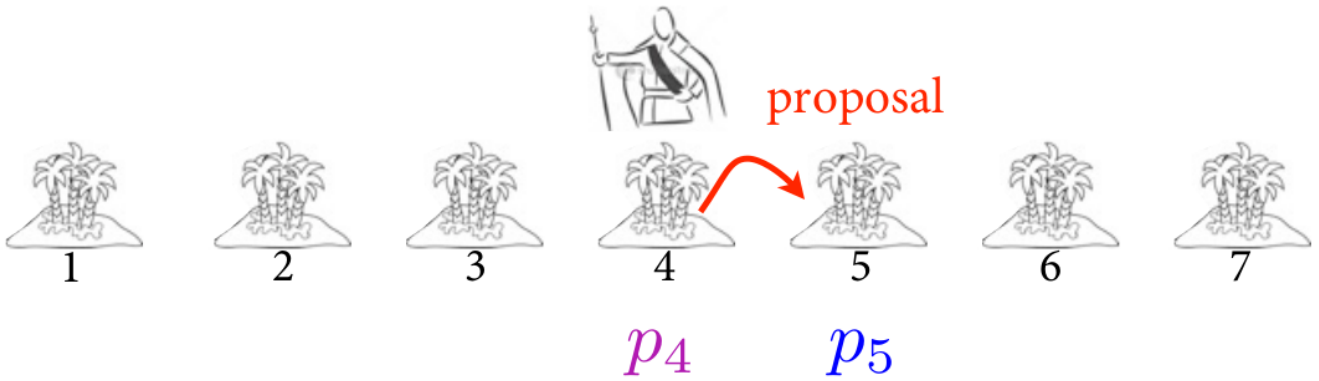


3. Find the population of the current island (p_4 here).

The Metropolis archipelago

Fair visit algorithm

1. small Flip a coin to choose an island left or right. Call it the "proposal" island.
2. Find the population of the proposal island (p_5 here).
3. Find the population of the current island (p_4 here).

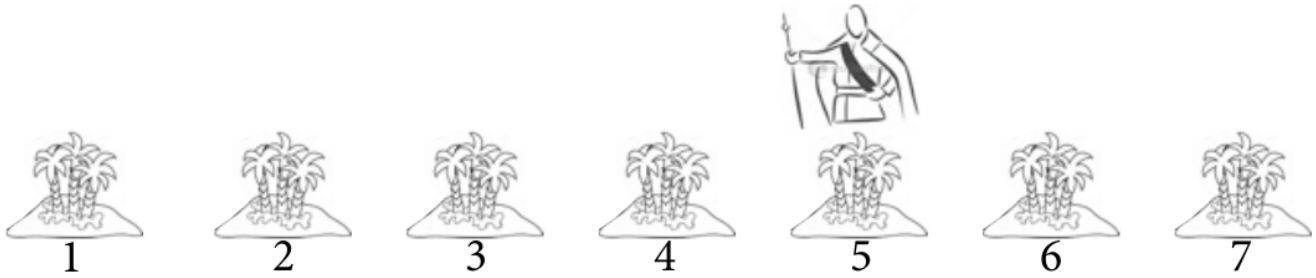


4. Move to the proposal with probability $\min(1, p_5/p_4)$.

The Metropolis archipelago

Fair visit algorithm

1. small Flip a coin to choose an island left or right.
   Call it the "proposal" island.
2. Find the population of the proposal island (p_5 here).
3. Find the population of the current island (p_4 here).
4. Move to the proposal with probability $\min(1, p_5/p_4)$.

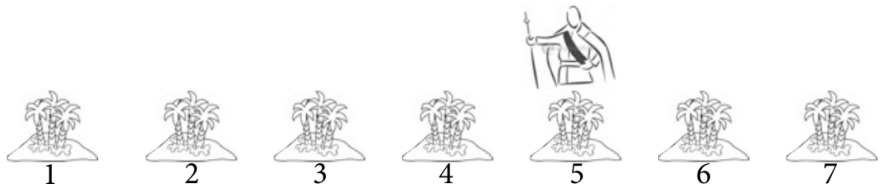


5. Repeat from 1.
   (with boundary condition: $1 \leftrightarrows 7$)

The Metropolis archipelago

Fair visit algorithm

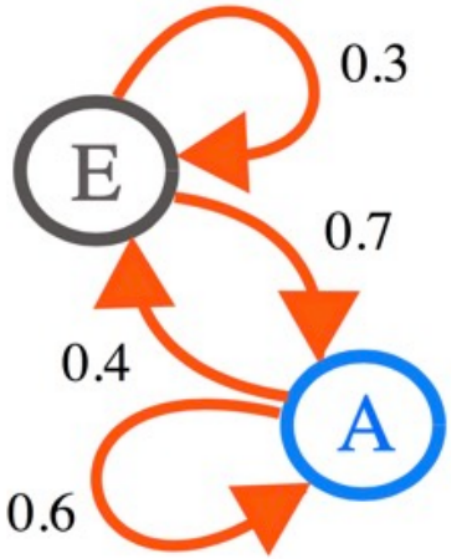
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2. Find the population of the proposal island (p_5 here).
3. Find the population of the current island (p_4 here).
4. Move to the proposal with probability $\min(1, p_5/p_4)$.
5. Repeat from 1.



This algorithm ensures that King Markov will be visiting each island in proportion to its population, in the long run.

Markov chain Monte Carlo

- Markov chain Monte Carlo
  - Understand the principle(s)
  - Meet different algorithms
- Interfaces to MCMC: `Stan` and `ulam`
  (and `BUGS`, Jags, `brms`, `rstandarm`, `blavaan`, etc.)
- How to sample sensibly?
- How to recognise and fix problems?



Metropolis "King Markov" version algorithm

```
num_weeks <- 100000 #This is more than 250 years!
positions(rep(0, num_weeks)
current <- 10
for(i in 1:num_weeks){
  #record current position
  positions[i] <- current

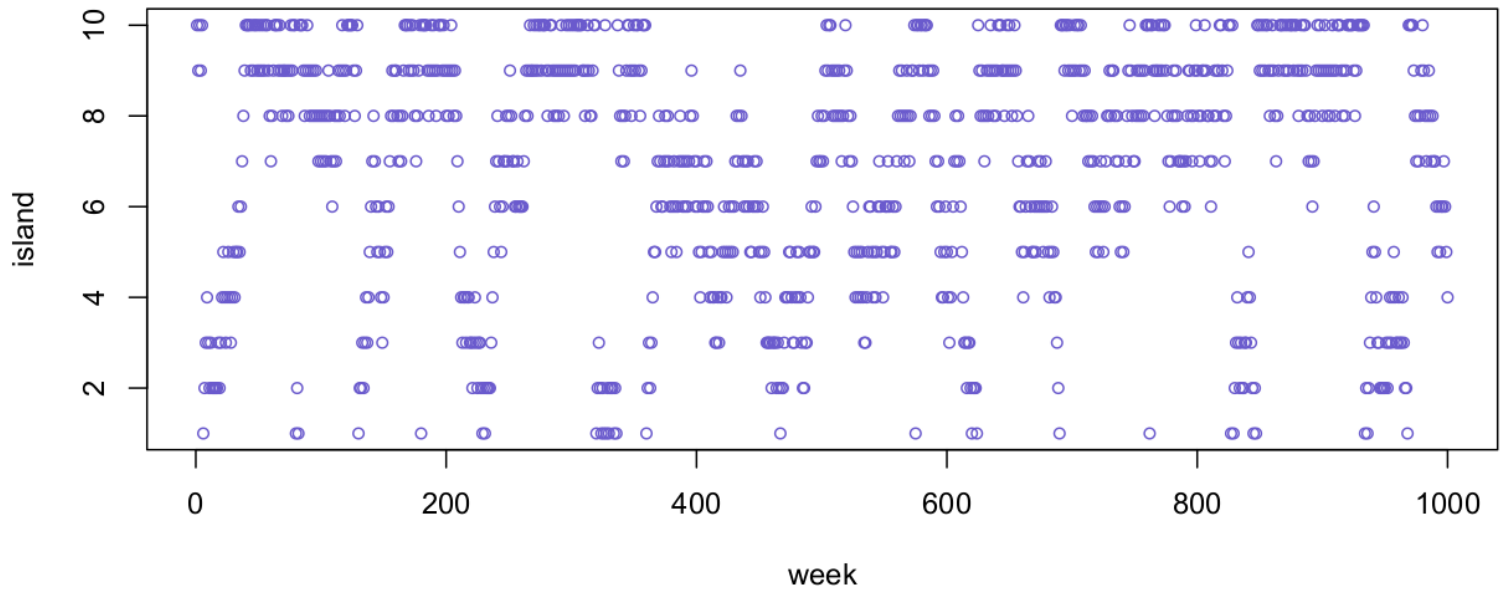
  #flip coin to generate proposal
  proposal <- current+sample(c(-1,1), size = 1)

  # Ensure loop around archipelago
  if(proposal<1){proposal <- 10}
  if(proposal>10){proposal <- 1}

  #move?
  prob_move <- proposal/current #omit min
  current <- ifelse(runif(1)<prob_move, proposal, current)
}
```

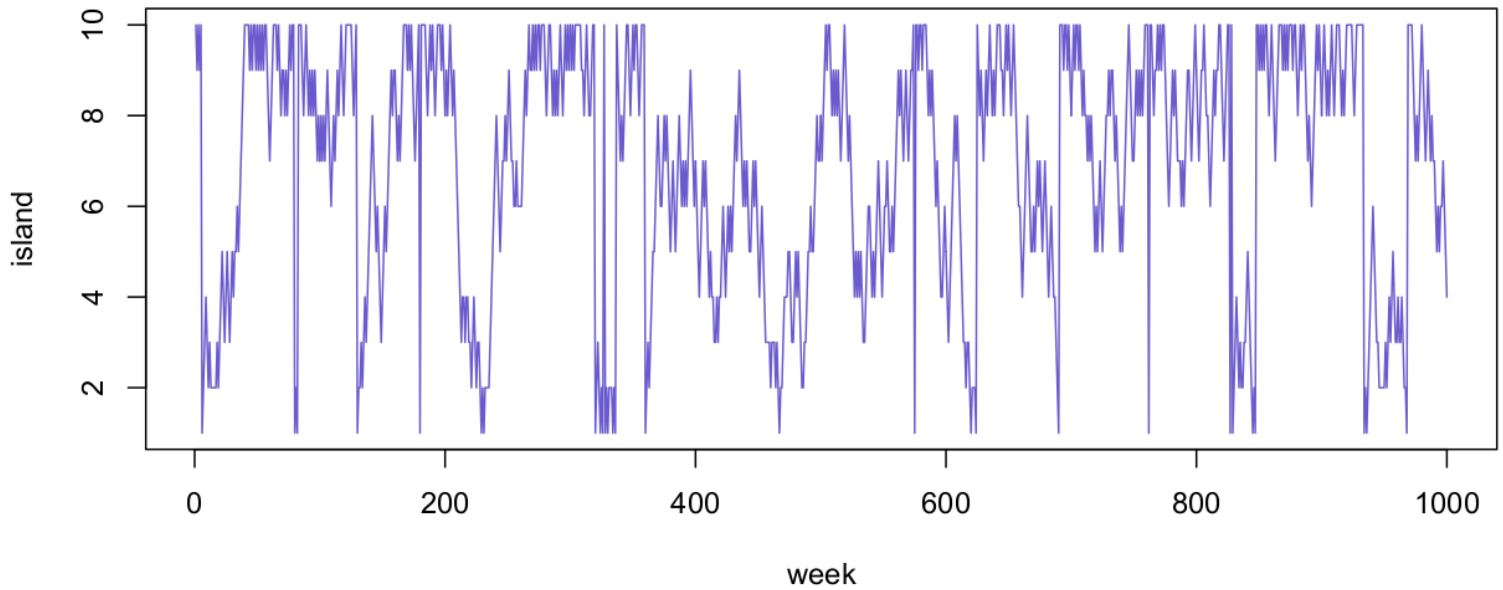
Metropolis "King Markov" version algorithm

Visit sequence



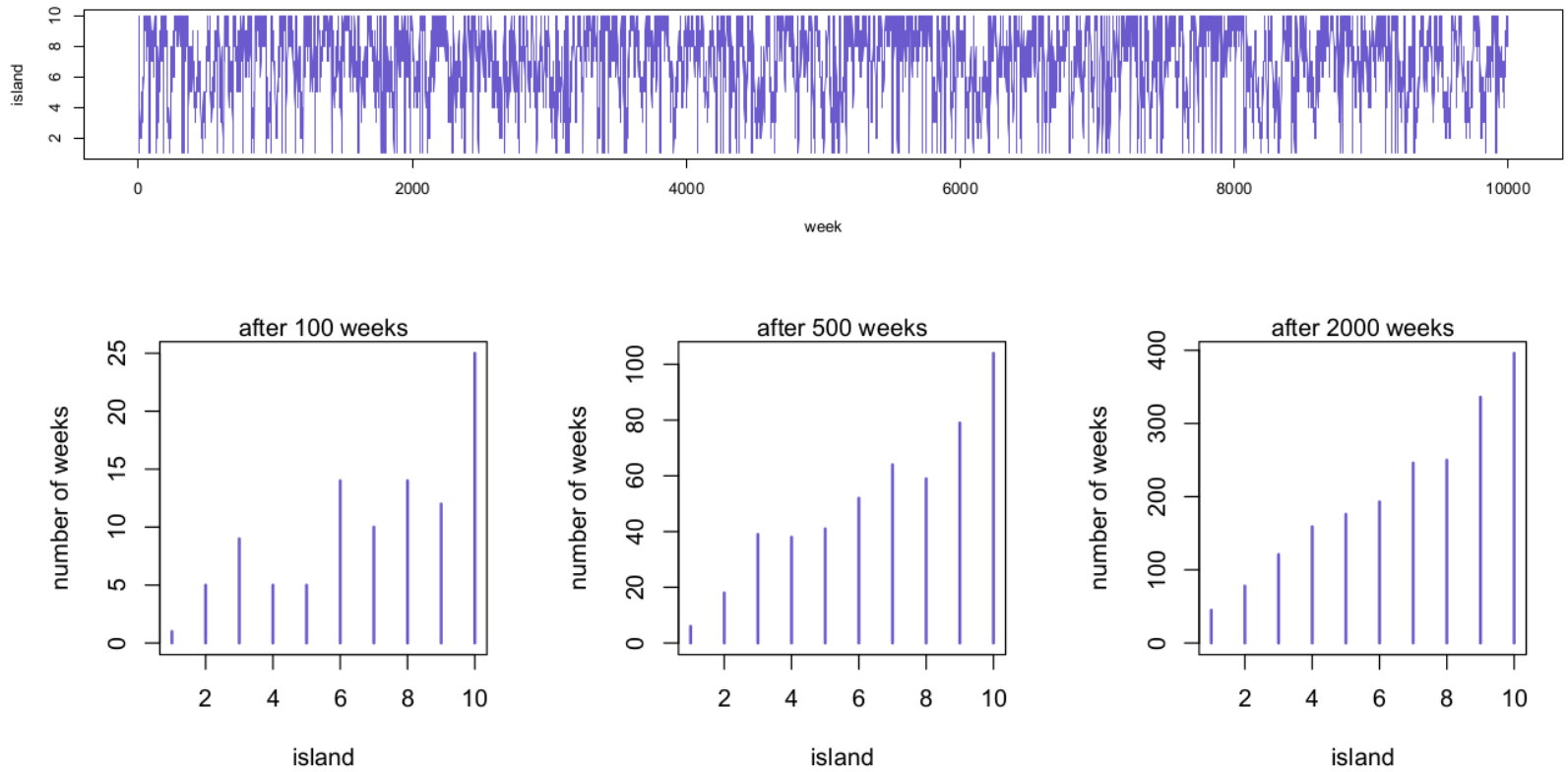
Metropolis "King Markov" version algorithm

Visit sequence



Metropolis "King Markov" version algorithm

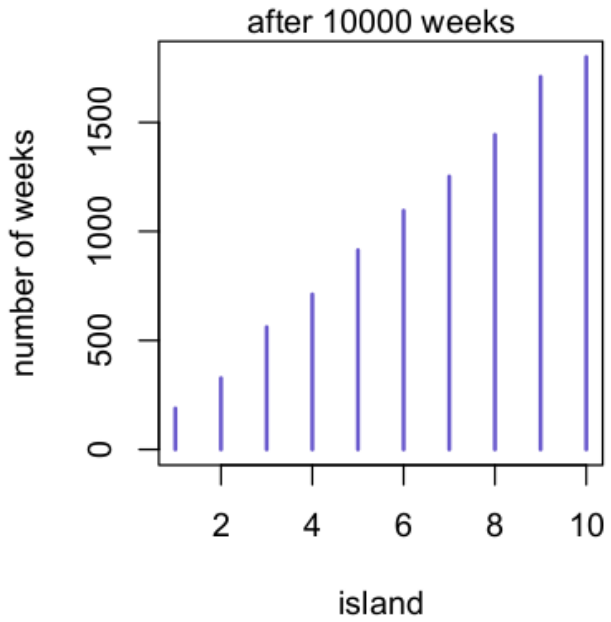
Limiting distribution



An equilibrium or stationary distribution?

King Markov sequential island visits

- Converges to correct proportions in the long run
- No matter which island is the starting one
- As long as proposal are symmetrical
- Key is in the acceptance condition
- This is an example of a *Metropolis algorithm*



Metropolis and MCMC

First ingredients

- Usual use is to draw samples from a posterior distribution
  - "Islands" = parameter values
  - "Population size" = proportional to posterior density for the parameter value
- Each parameter value is visited in proportion to its posterior probability. Even without having it entirely characterised!?!
- Works for any number of dimensions (# parameters)...in theory.
- Works for continuous or discrete parameter space.





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Metropolis and MCMC

History

- Metropolis algorithm is a simple example of an MCMC algorithm
- Metropolis, Rosenbuth, Rosenbluth & Teller, 1953.

THE JOURNAL OF CHEMICAL PHYSICS VOLUME 21, NUMBER 6 JUNE, 1953

Equation of State Calculations by Fast Computing Machines

NICHOLAS METROPOLIS, ARIANNA W. ROSENBLUTH, MARSHALL N. ROSENBLUTH, AND AUGUSTA H. TELLER,
Los Alamos Scientific Laboratory, Los Alamos, New Mexico

AND

EDWARD TELLER,* *Department of Physics, University of Chicago, Chicago, Illinois*

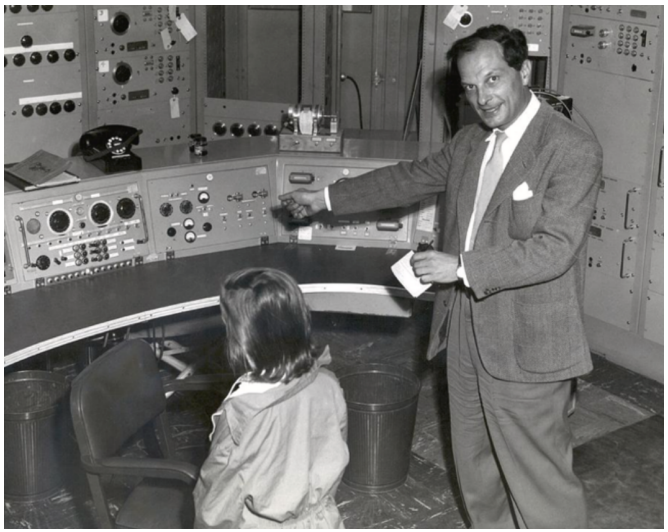
(Received March 6, 1953)

A general method, suitable for fast computing machines, for investigating such properties as equations of state for substances consisting of interacting individual molecules is described. The method consists of a modified Monte Carlo integration over configuration space. Results for the two-dimensional rigid-sphere system have been obtained on the Los Alamos MANIAC and are presented here. These results are compared to the free volume equation of state and to a four-term virial coefficient expansion.

Metropolis and MCMC

MANIAC

- MANIAC: Mathematical Analyzer, Numerical Integrator and Computer
  - $1,000$ pounds,
  - 5 kilobytes of memory and
  - $70k$ multiplications/sec.
- Your laptop
  - $4 - 7$ pounds,
  - $4 - 32$ million kilobytes of memory and
  - Billions of multiplications/sec.



Stanislaw Ulam, his daughter Claire and MANIAC
Ariana Rosenbluth made the maths work in the machine

Metropolis and MCMC

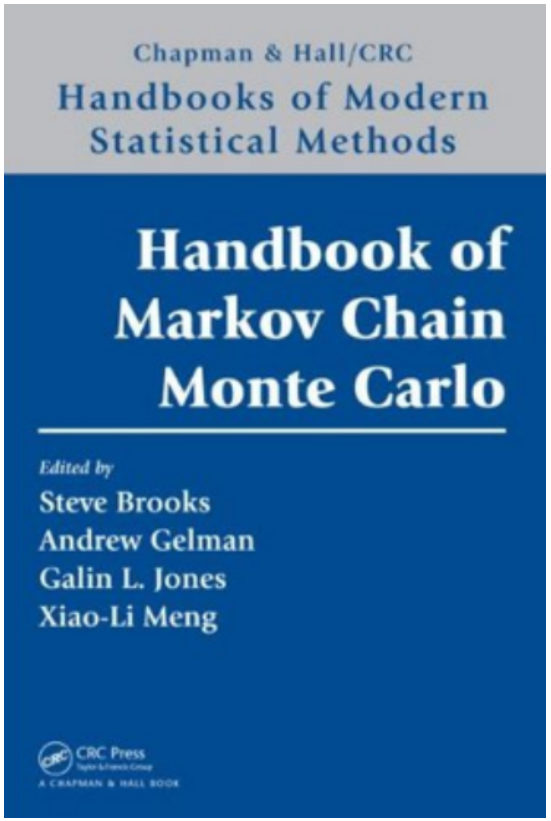
More ingredients; why?

- Metropolis algorithm is a simple example of an MCMC algorithm. Easy to write, but not that efficient.
- *Chain* = sequence of draws from the target distribution
- Markov chain: the philosophy is that the past of the chain does not matter, only where you are now and the evolution rule(s).
- Monte Carlo: because we do random simulations.

- Sometimes, we cannot write an integrated posterior. Even if we could, can we use it?
- This is the case in many applications fields: multi-level models, networks, phylogenies, spatial models, etc.
- Direct optimisation is a poor strategy in high-dimensions. MCMC is not fancy, it is required. And seasoned now.

MCMC strategies

- Metropolis: ancestor of all MCMC algorithms
- Metropolis-Hastings (M-H) is more general
- Gibbs sampler (GS) is an efficient version of the M-H algorithm
- Both M-H and GS are "guess and check" strategies.
- Hamiltonian Monte Carlo (HMC, aka Hybrid Monte Carlo) is fundamentally different
- New methods are always developed, but the future belongs to gradient approaches.



Online demo

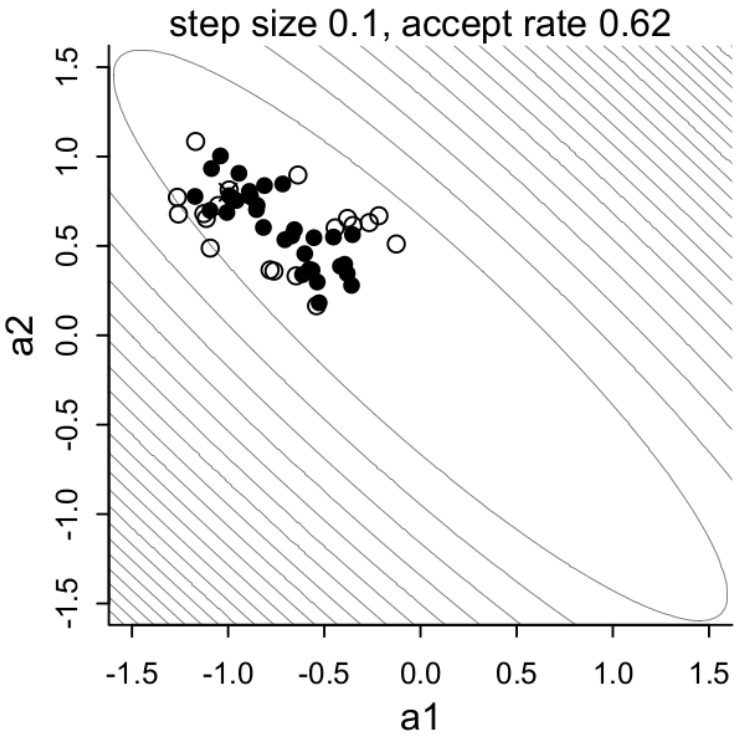
https://chi-feng.github.io/mcmc-demo/

Chi Feng's "Markov-chain Monte Carlo Interactive Gallery"

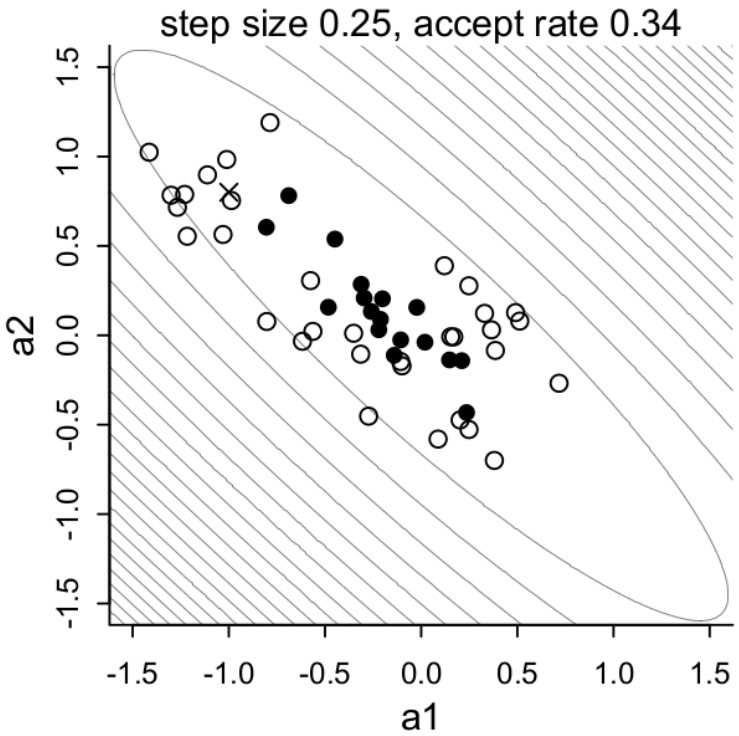
Try simple shape first and then get exotic (multimodal, "ridge" situation, etc.)

Metropolis gets stuck

Even in a moderate condition scenario



small steps, slow walk



bigs steps, low accept rate

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